

CONNECTOR

Architecture Record — Agent Operating Substrate for Augmented Environments

Complete technical record of what has been built: system architecture, isolation stack (AgentCell · MicroCell · CVR · Firecracker), engineer and agent capabilities, product technicalities, honesty bounds, how to operate, and outcomes delivered.

Document: `platform/docs/arch/arch.pdf`

Status: Architecture / product record (capability + honesty, not absolute security)

Companions: CONNECTOR_CAPABILITY_STANDARD · PRODUCT_PROMISE · FINAL_OUTCOMES · AGENT_ISOLATION · OPERATIONAL_SOFTWARE · REACH_CHECKLIST

Generated from engineered surfaces in `platform/server`, `platform/microvm`, `platform/connector-microd`

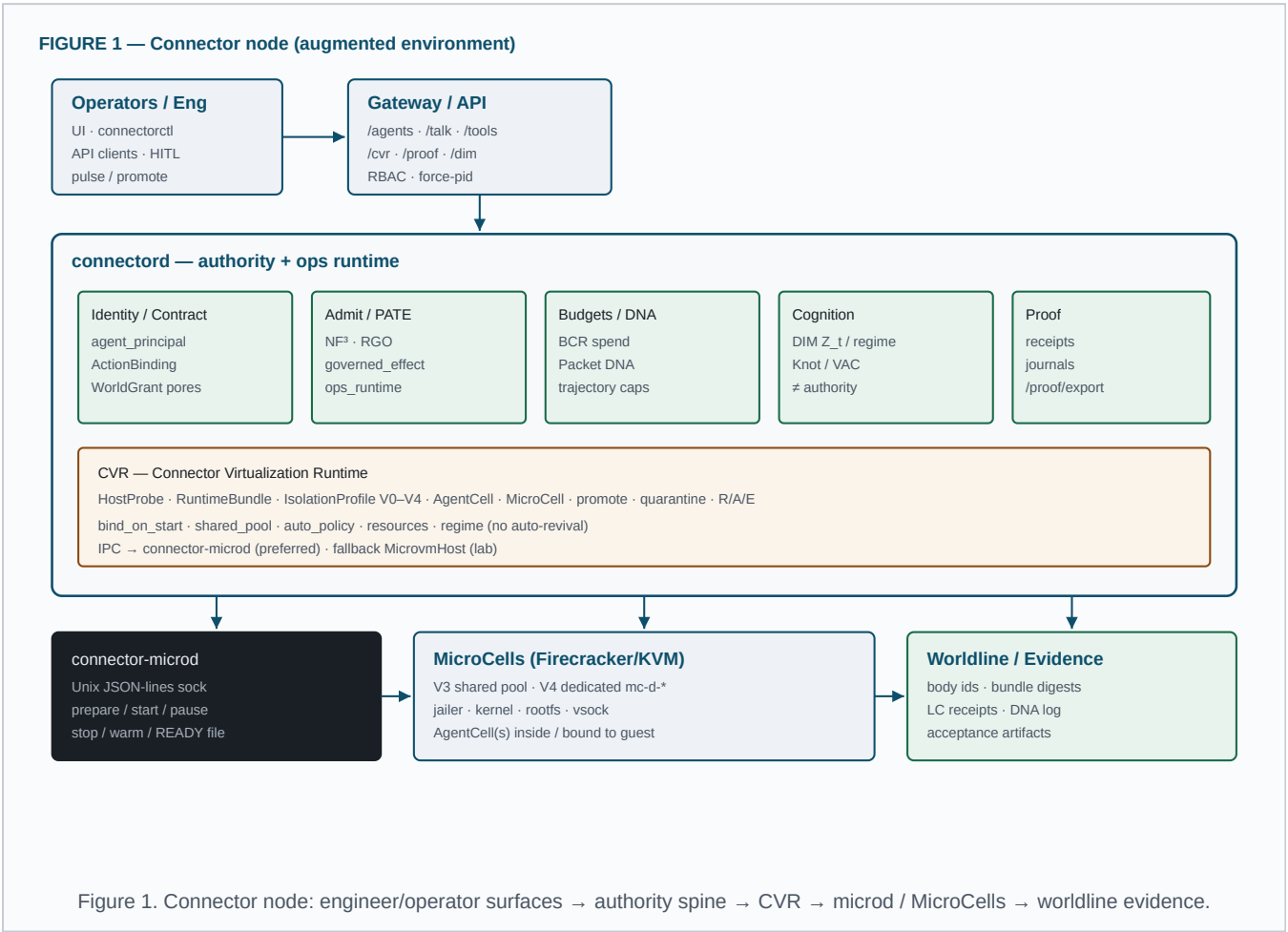
Honesty frame. Connector is engineer tooling for hosting dynamic agents under chosen constraints. It does *not* claim “100% secure,” SIL/physical safety, or organizational compliance certification. It claims: explicit posture, applied gates, observable denial, reconstructible worldline.

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2. Complete architecture drawing (system)

Figure 1 is the end-to-end Connector node: operator & engineer surfaces, connectord authority spine, CVR isolation plane, guest MicroCells, evidence export, and optional TraceTramp / Witness paths.

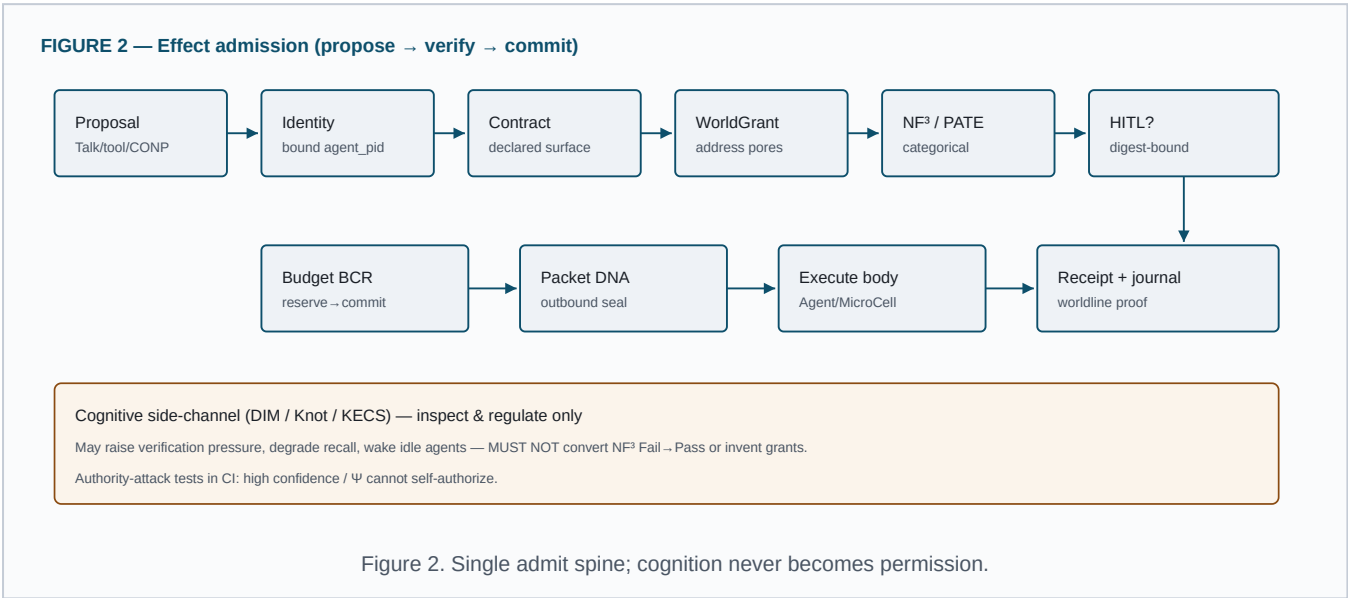


2.1 Data & control flows (summary)

Flow	Path	Gate
Talk	Gateway → ops_runtime preflight → PATE → BCR → recall → model	identity, contract, budgets, frozen deny
Tool / MCP	preflight → knowledge justify → DNA → governed_effect → PATE	WorldGrant, ActionBinding
Agent start	assert_harden_ready → cvr::bind_on_start → AgentCell ± MicroCell	START_REFUSED if required unmet
Promote	POST /agents/:pid/isolation/promote → same agent_pid, new body	HostProbe / microd
Quarantine	deny → inflight → grants → net → freeze → VMM → verify → progeny	QUARANTINE_FAILED on miss
Proof	GET /proof/export/:pid + connectorctl worldline export	integrity sha256 + posture triad

3. Authority & effect spine

Every mutating consequence is meant to pass a single conceptual admit path. Soft/playground may label LAB; harden must refuse or fail closed.



3.1 Effect exclusivity

When `CONNECTOR_EFFECT_EXCLUSIVITY=1` (on under augmented env / production presets), alternate mutating paths (raw shell, ungated child spawn, CONP bypass) are closed or refused. Ungated paths are bugs to close, not features.

4. Agent isolation architecture (three layers)

Thesis: Connector isolates *agents*, not merely processes. Three layers are required and composed.



4.1 Engineer isolation surface

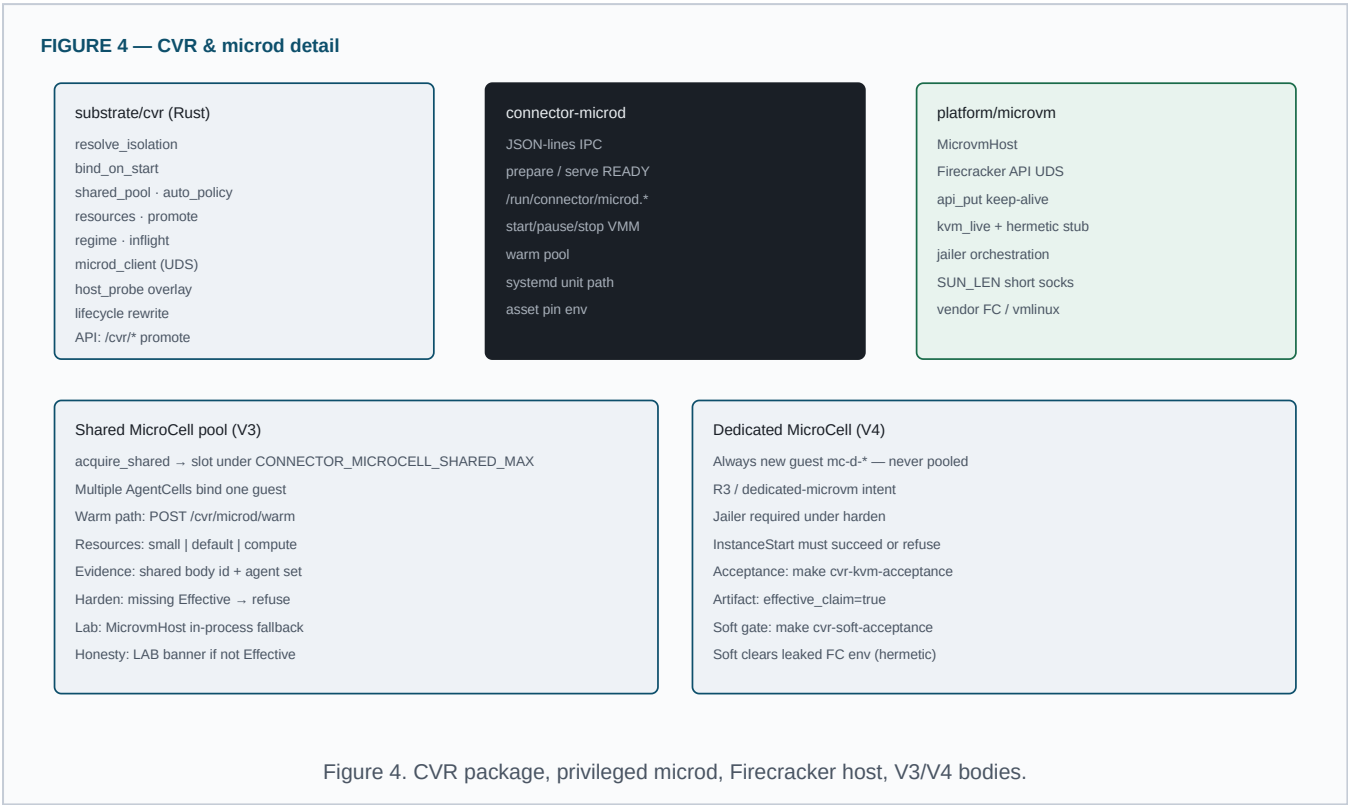
```
agent:
  isolation: auto          # linux-cell | hardened-linux-cell | shared-microvm | dedicated-
microvm
  isolation_risk: R2       # drives auto table only — never grants authority
  resources: default      # small | default | compute
```

4.2 Profiles V0–V4

Profile	Intent	Body	Typical risk map (auto)
V0	App / contract gates only	process / soft cage	lab / R0 exploratory
V1	Linux cell	AgentCell (ns/cgroup)	R0 → V1
V2	Hardened Linux cell	AgentCell + Landlock fail-closed	R1 → V2
V3	Shared MicroCell	N AgentCells / one Firecracker guest	R2 → V3
V4	Dedicated MicroCell	one guest per agent (mc - d - *)	R3 → V4

Posture triad on every isolation decision: **Requested / Applied / Effective**. Missing KVM/assets/jailer when required → `START_REFUSED`, never silent “applied.”

5. CVR · MicroCell · Firecracker · connector-microd



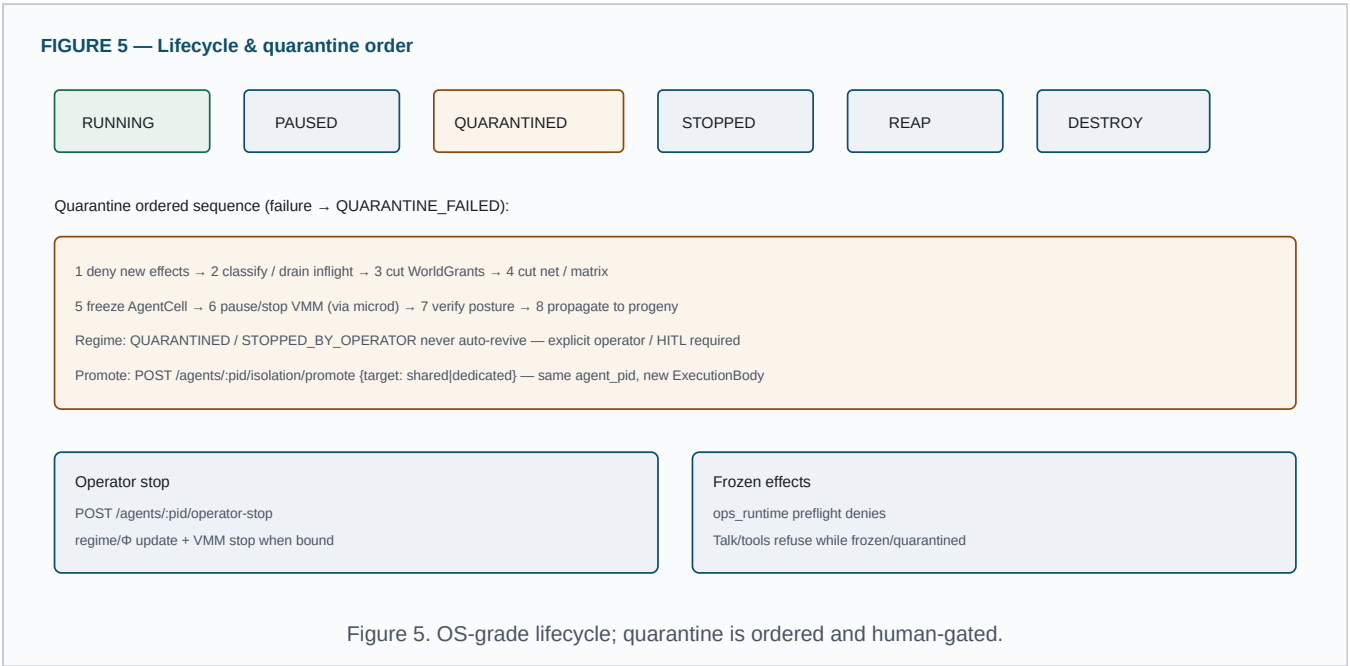
5.1 HostProbe & Effective

Signal	Meaning
Requested	Engineer/config asked for a profile (e.g. dedicated-microvm)
Applied	Runtime attempted bind with available controls
Effective	Host can actually run Firecracker MicroCell (KVM + kernel + rootfs + start proven)
LAB / unavailable	Honest when Effective unmet — never marketed as harden-applied

5.2 Asset pins

```
export CONNECTOR_FIRECRACKER_BIN=/path/to/firecracker
export CONNECTOR_JAILER_BIN=/path/to/jailer # required under harden
export CONNECTOR_MICROVM_KERNEL=/path/to/vmlinux
export CONNECTOR_MICROVM_ROOTFS=/path/to/rootfs.ext4
export CONNECTOR_MICROVM_STATE_DIR=/var/lib/connector/microvm/instances
export CONNECTOR_MICROD_SOCK=/run/connector/microd.sock
# systemctl enable --now connector-microd.service
```

6. Lifecycle, promote, quarantine, regime



7. Augmented environment enablement

A *real* augmented environment node (not lab) flips membrane defaults and harden refuse-start:

```
export CONNECTOR_AUGMENTED_ENV=1
# or CONNECTOR_PRESET=production / CONNECTOR_ENV=production
```

Flag	Role
CONNECTOR_HARDEN_REFUSE_START=1	Start refuses when mandatory gates unmet
CONNECTOR_EFFECT_EXCLUSIVITY=1	Alternate effect paths closed
CONNECTOR_SANDBOX_UNBYPASSABLE=1	FS/net/VM/broker bar
CONNECTOR_ISOLATION_RUNTIME=microvm	Isolation grade for exclusivity

Inspect: GET /api/v1/substrate/status → ops_runtime + harden_posture + cvr + escape_hatches + lab_banner .

Playground (CONNECTOR_PLAYGROUND=1) never engages augmented harden refuse-start.

8. Forty capability claims (agentic augmented environment)

These are product claims Connector asserts it can provide when configured. Each is capability + honesty — not a warranty of absolute security.

01 Bound agent identity — no ambient LLM effects under harden.	02 Contract floor — undeclared FS/net/tools denied.
03 WorldGrant pores — external addresses only through grants.	04 Packet DNA on outbound effects when required.
05 Continuity Broken → egress cut (matrix / governed_effect).	06 Memory namespace <code>m/{kernel_pid}</code> per agent.
07 Ingress tokenization / seal via LLM broker when enabled.	08 Detokenize only on admitted effects.
09 BCR token/compute budgets with reserve → commit.	10 Trajectory blast / commitment / exposure caps.
11 Persistent inspectable DIM cognitive state Z_t .	12 DIM regulates cognition without minting Allow.
13 Durable VAC memory + recall across restart.	14 Knot interference on contradiction (no silent overwrite).
15 Mission continuity / idle wake without new prompt.	16 Model-replaceable mind (identity + DIM persist).
17 Digest-bound HITL for high-risk actions.	18 Proportional autonomy R0–R3 via RGO/PATE.
19 Receipt for every admitted effect.	20 Compensation when inverse registered (AAPI).
21 CONP without SIL theater (honest partner HAL).	22 Narrowed child agents with paired receipts.
23 Operator stop/inspect without raw chain-of-thought.	24 Authority-attack resistance (cognitive ≠ permission).
25 Isolation continuum V0–V4 including MicroCell.	26 Shared MicroCell pool for density (V3).
27 Dedicated MicroCell for blast radius (V4).	28 <code>isolation: auto</code> risk → profile table (configurable).
29 Promote body without new agent_pid.	30 Ordered quarantine with QUARANTINE_FAILED honesty.
31 No auto-revival from QUARANTINED / operator-stop.	32 START_REFUSED when Requested gates unmet (augmented).
33 Posture triad Requested/Applied/Effective on CVR.	34 Lab honesty — soft-fail never sold as harden.
35 Named escape hatches + audit in substrate status.	36 Proof export (digests, DNA, journals, integrity hash).
37 Operator pulse: denials, spend, HITL, DIM.	38 Effect exclusivity when membrane applied.
39 Playground → harden continuum on one spine.	40 KVM Effective acceptance artifact for MicroCell claim.

Mapped to E1–E8 · A1–A28 · S1–S28 in CONNECTOR_FINAL_OUTCOMES.md and REACH checklist.

9. Why an engineer should use it

Problems it solves

- Agents that “just call tools” with ambient authority
- No way to prove what was admitted vs what the model said
- Lab demos silently treated as production-hardened
- Isolation knobs scattered (cgroup here, prompt rules there)
- No promote path when risk rises mid-life
- Cognitive confidence mistaken for permission

What you gain

- One envelope language: contract · grant · tier · budget · HITL
- Same APIs from playground to harden
- OS-grade lifecycle + MicroCell when you need it
- Exportable worldline for incident / audit review
- Explicit LAB banners when Effective is unmet
- DIM/Knot for cognition without authority bleed

Engineer job. Choose the point on the dynamic ↔ constrained continuum; Connector makes that choice real, observable, and reconstructible.

10. Capabilities the agent receives

When hosted on Connector (gates enabled), an agent gains a *bounded* operating environment — not unrestricted host power.

Capability	What the agent gets	Bound
Talk / models	Routed inference under identity	budgets, tokenization, NF ³
Tools / MCP	Declared tool surface	contract + PATE + DNA
Memory	Private namespace + VAC recall	knowledge boundary / DIM radius
World addresses	Only granted pores	WorldGrant fail-closed option
Mission	Journalled continuity / wake	trajectory caps
Children	Narrowed spawn with receipts	parent envelope inheritance
CONP	Protocol effects with DNA	no SIL theater from stubs
Isolation body	AgentCell / MicroCell as configured	cannot self-promote Effective

The agent does **not** get: ambient root, silent grant expansion, self-authorization via confidence, or auto-revival after quarantine.

11. Product & technical inventory

11.1 Major code surfaces

Area	Location	Role
Server / API	platform/server	connectord, ops_runtime, gateway
CVR	platform/server/src/substrate/cvr/	isolation resolve, microd client, pool, promote, regime
MicroVM	platform/microvm	Firecracker host, kvm_live, hermetic stub
microd	platform/connector-microd	Privileged VMM supervisor IPC
Acceptance	scripts/cvr-*-acceptance.sh	Soft LAB + KVM Effective gates
CI	.github/workflows/cvr-kvm-acceptance.yml	Soft always; KVM-gated Effective
UI	platform/ui-leptos	Operator dashboard surfaces
Docs SoT	platform/docs/arch/*	Capability, promise, outcomes, isolation

11.2 Technical properties

- Rust async server; Unix domain socket JSON-lines to microd
- Firecracker API over short UDS paths (SUN_LEN); HTTP keep-alive body drain
- Resource profiles map to guest sizing; shared max configurable
- Auto policy JSON overrides default R → V table
- Vendor manifests pin SHA256 for FC/kernel/rootfs (x86_64)
- Large VMM binaries gitignored; acceptance uses measured assets

12. Tool — not full security software

Connector is a **governed-autonomy toolkit / agent OS substrate**. It is not a replacement for enterprise GRC suites, endpoint EDR, network firewalls, physical SIL, or legal compliance programs.

What people might assume	What Connector actually is
"Install Connector → agents are secure"	Engineer must choose and apply posture; misconfig stays soft/LAB
"We are HIPAA/SOC2 certified by Connector"	Evidence/export for org mapping; optional GRC routes ≠ certification
"LLM cannot be jailbroken"	Membrane gates consequence; model content attacks remain a risk class
"MicroCell = airgap"	KVM boundary + semantic gates; covert channels / host compromise out of scope
"Partner HAL/SIL proven"	Physical safety stays partner-attested — never claimed from lab stubs
"DIM high confidence = safe to act"	Confidence never becomes Allow

12.1 Correct use of the tool

1. Define agent contract and WorldGrants explicitly.
2. Pick isolation profile (or auto + risk) matching blast radius.
3. Enable augmented env / harden for production nodes.
4. Verify Effective on hosts that claim MicroCell.
5. Monitor via pulse; quarantine with human gate when needed.
6. Export worldline for incidents — do not invent "compliance green."

13. What we built (accomplishment record)

13.1 Product / honesty spine

- Capability standard, product promise, E/A/S outcomes, REACH checklist
- Anti-claim audit script in CI (`audit-product-promise.sh`)
- Augmented env flag + refuse-start membrane

13.2 Authority & agent capabilities

- Identity, contract, WorldGrant, Packet DNA, BCR spend on live Talk/tool paths
- DIM / Knot / VAC / mission journal with authority-attack tests
- Proof export + operator pulse without CoT
- Effect exclusivity and escape-hatch enumeration

13.3 Isolation Phases A–F (CVR)

Phase	Delivered
A	V0–V4 vocabulary, ExecutionBody kinds, R/A/E on isolation, START_REFUSED mapping
B–C	AgentCell / MicroCell binding, lifecycle toward VMM, evidence hooks
D	connector-microd prepare/serve READY, IPC client, status/warm APIs
E	Shared pool, dedicated mc-d-*, auto_policy, resource profiles
F	Promote same pid, ordered quarantine, regime no auto-revival

13.4 Linux-level Effective acceptance

- `make cvr-soft-acceptance` — always; LAB honesty / refuse-start
- `make cvr-kvm-acceptance` — KVM host; writes `effective_claim=true` artifact on PASS
- Firecracker keep-alive HTTP fix, short socket paths, hermetic soft env clearing

14. How to use (operator runbook)

14.1 Dev / playground

```
# soft exploration
export CONNECTOR_PLAYGROUND=1
# start connectord per BOOTSTRAP_RUNBOOK / Makefile
# inspect: GET /api/v1/substrate/status → expect LAB labels when gates off
```

14.2 Augmented / production posture

```
export CONNECTOR_AUGMENTED_ENV=1
systemctl enable --now connector-microd.service # if MicroCell required
# pin FC/jailer/kernel/rootfs (see §5.2)
# register agent with isolation: auto|shared-microvm|dedicated-microvm
# start: POST /agents/:pid/start – expect START_REFUSED if Effective unmet
```

14.3 Day-2 ops

```
# status
curl -s $BASE/api/v1/cvr/status | jq .
curl -s $BASE/api/v1/operator/pulse | jq .

# promote
curl -s -X POST $BASE/api/v1/agents/$PID/isolation/promote \
  -H 'content-type: application/json' \
  -d '{"target": "dedicated"}'

# quarantine / stop
curl -s -X POST $BASE/api/v1/agents/$PID/quarantine
curl -s -X POST $BASE/api/v1/agents/$PID/operator-stop

# proof
curl -s $BASE/api/v1/proof/export/$PID -o worldline.json
connectorctl worldline export --agent $PID
```

14.4 Acceptance on a host

```
make cvr-soft-acceptance
make cvr-kvm-acceptance # requires /dev/kvm + assets
# artifact: artifacts/cvr-acceptance/cvr-kvm-acceptance.json
```

15. Outcomes & acceptance evidence

Outcome set	IDs	Meaning
Engineer freedom	E1–E8	Configure and prove posture continuum
Agent capabilities	A1–A28	What hosted agents gain / how limits bind
System capabilities	S1–S28	Platform working software bands

Pass rule. Capability + honesty bars — not “100% secure” certification. Soft postures must be labeled soft. Harden must fail closed or refuse start.

Primary verify surfaces: `/substrate/status`, `/product/promise`, `/operator/pulse`, `/proof/export`, `/cvr/status`, soft + KVM acceptance artifacts, `audit-product-promise.sh`.

15.1 Outcomes this stack delivers

- Engineers can dial isolation and budgets without rewriting the agent identity.
- Operators can freeze, quarantine, and stop including VMM when bound.
- Auditors can reconstruct admitted effects from exportable evidence.
- Hosts without KVM cannot claim MicroCell Effective.
- Labs remain useful without polluting harden claims.

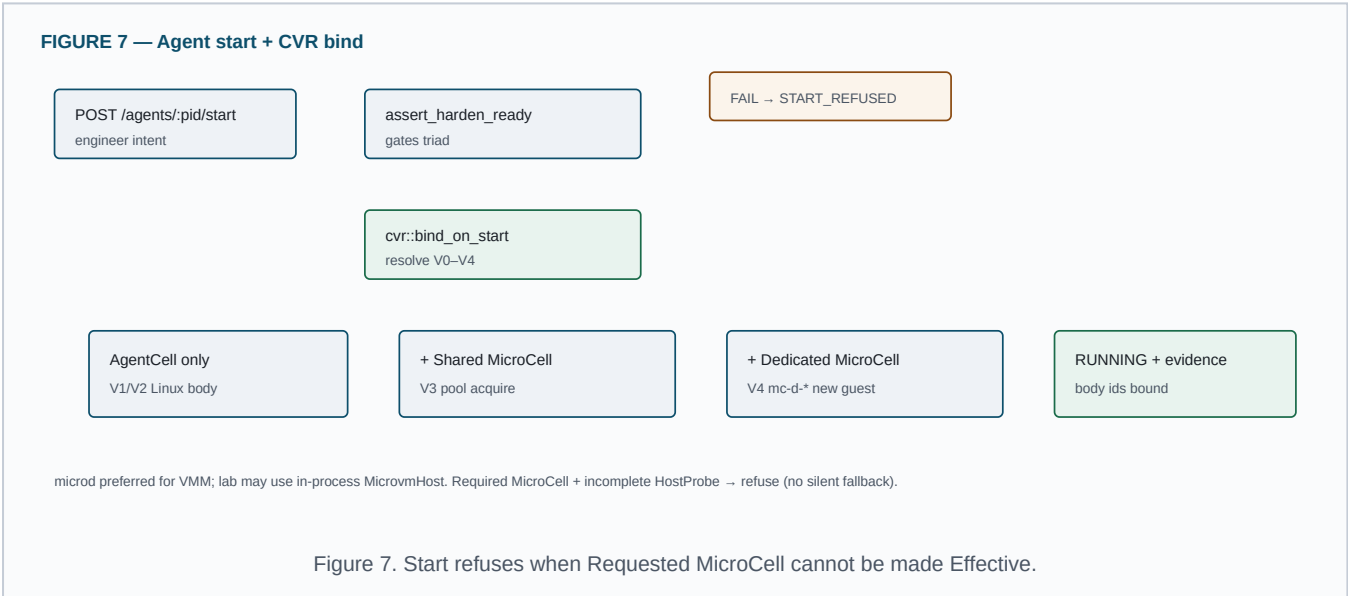
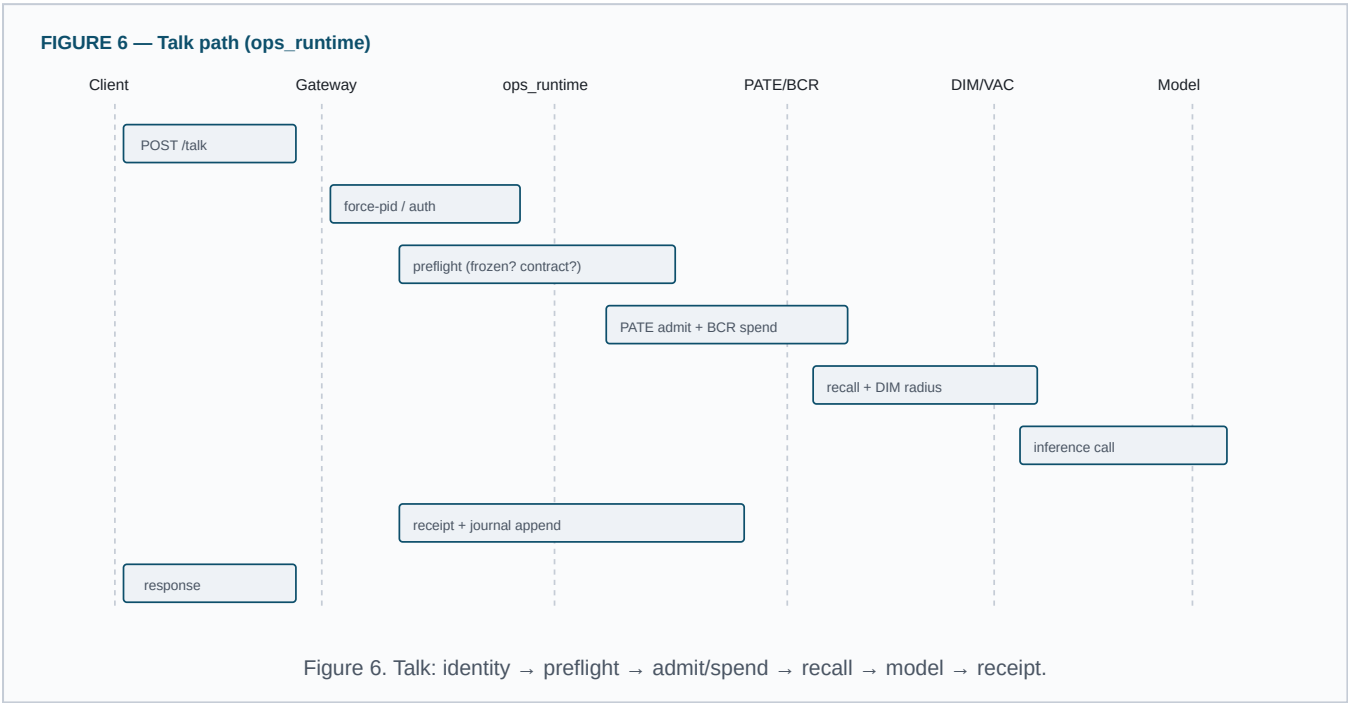
16. Anti-claims & known limits

Anti-claim	Why
100% secure	Covert channels, host compromise, novel LLM bypasses, misconfig
Every deploy equally hard	Playground soft-fail is intentional entropy leak
Cognition is correct	We gate consequence; we do not certify model truth
Lab echo is SIL	Physical safety is partner / attested path only
Confidence = permission	DIM/Knot/KECS never mint Allow
We make an org compliant	Compliance is jurisdiction/process; we supply controls + evidence

Partner HAL remains LAB where not host-attested. T4 MicroCell Effective requires host proof.

17. Deep dive — Talk / tool / start sequences

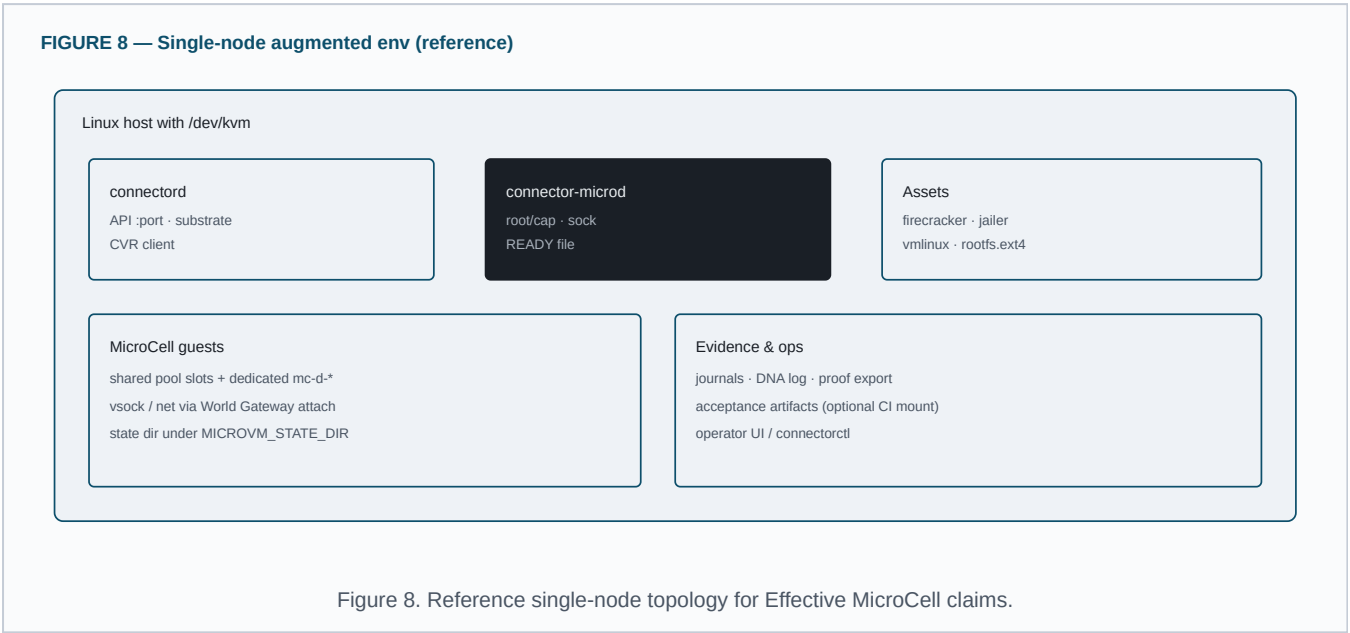
These sequence diagrams encode the live operational paths summarized in CONNECTOR_OPERATIONAL_SOFTWARE.



17.1 Failure modes (engineered honesty)

Condition	Response	Must not claim
Harden gate missing	START_REFUSED	Applied harden
KVM/assets absent for V3/V4 required	Refuse or LAB unavailable	Effective MicroCell
Quarantine step fails	QUARANTINE_FAILED	Silent partial quarantine success
Soft-fail Landlock	soft_fail_lab posture	Production-ready cage
Playground session	Soft membrane	Augmented env harden

18. Deployment topologies



18.1 Multi-agent density vs blast radius

Goal	Profile	Trade-off
Many low-risk agents	V1/V2 AgentCell or V3 shared	Density; shared guest fate
High-risk single agent	V4 dedicated	Cost/latency; strong blast radius
Risk rises mid-life	Promote shared → dedicated	Same pid; brief pause window
Unknown / mixed fleet	isolation: auto + risk	Policy table must be reviewed

19. Evidence model & worldline

Proof is a first-class product layer. Connector does not certify that “the agent was good”; it certifies what was *admitted*, what body executed, and what posture was Requested/Applied/Effective.

Artifact	Contents	API / tool
Action digests	Canonical action identity for HITL	ActionBinding
Receipts	Per admitted effect	AAPL bridge / mission journal
Packet DNA log	Outbound seals	DNA require mode
DIM journal	Cognitive condition history	/dim · export
Lifecycle receipts	Pause/quarantine/stop + VMM	CVR lifecycle
Bundle digests	Measured FC/kernel/rootfs	RuntimeBundle / HostProbe
Worldline export	Integrity sha256 + posture triad	/proof/export · connectorctl
Acceptance JSON	effective_claim for MicroCell	cvr-kvm-acceptance

Org compliance mapping (HIPAA/SOC2/EU AI Act report UIs) may exist as optional tooling; they are **out of scope** for “standard reached.” The substrate finish line is evidence the org can map themselves.

20. Comparison: what Connector is vs adjacent categories

Category	Typical product	Connector relation
LLM app framework	Chains, prompts, tool routers	Connector hosts agents under membrane; not only a prompt SDK
Container orchestrator	k8s pods	AgentCell/MicroCell are agent-centric bodies, not generic schedulers
Secret manager / IAM	Keys, roles	Uses identity + grants; does not replace org IAM
SIEM / EDR	Host threat detection	Exports worldline; does not replace SOC tooling
GRC platform	Control frameworks	Supplies controls+evidence; not certification
Hypervisor product	VMware / cloud VMs	Uses Firecracker as backend behind MicroCell UX

Positioning: **agent operating substrate for augmented environments** — closest analogy is “OS + cgroups + audit for agents,” not “antivirus for AI.”

21. Appendix A — Forty claims (expanded verify hints)

#	Claim	Verify hint
1	Bound identity	Anon Talk banned under harden; force-pid
2	Contract floor	Undeclared tool → deny
3	WorldGrant pores	Ungranted URL → deny
4	Packet DNA	DNA log entry on tool egress
5	Continuity cut	Broken continuity → egress deny
6	Memory ns	Packets only under <code>m/{pid}</code>
7–8	Tokenize / detokenize	Broker gate on; detok only admitted
9–10	Budgets / trajectory	Exhaust → refuse spend/effects
11–12	DIM persist / non-authority	Restart <code>Z_t</code> ; authority-attack CI
13–14	VAC / Knot	Recall after restart; interference API
15–16	Wake / model swap	wake/evaluate; <code>set_model_ref</code>
17–18	HITL / R0–R3	Digest approval; R3 fail-closed
19–20	Receipts / compensate	AAPi ledger + inverse
21–22	CONP / children	DNA path; paired child receipts
23–24	Operator / anti self-auth	pulse no CoT; audit-authority-attack
25–28	V0–V4 / auto	isolation resolve + policy JSON
29–31	Promote / quarantine / regime	promote API; ordered QC; no revive
32–35	Refuse / R-A-E / LAB / hatches	substrate status triad + <code>escape_hatches</code>
36–37	Proof / pulse	<code>export sha256</code> ; pulse fields
38–39	Exclusivity / continuum	exclusivity flag; <code>playground#harden</code>
40	KVM Effective	<code>effective_claim=true</code> artifact

22. Appendix B — Environment reference (selected)

Variable	Purpose
CONNECTOR_AUGMENTED_ENV	Production membrane + refuse-start defaults
CONNECTOR_PLAYGROUND	Soft lab posture
CONNECTOR_HARDEN_REFUSE_START	Start refuse when gates unmet
CONNECTOR_EFFECT_EXCLUSIVITY	Close alternate mutate paths
CONNECTOR_SANDBOX_UNBYPASSABLE	FS/net/VM/broker bar
CONNECTOR_ISOLATION_RUNTIME	Isolation grade (e.g. microvm)
CONNECTOR_MICROCELL_SHARED_MAX	V3 pool capacity
CONNECTOR_ISOLATION_AUTO_POLICY_JSON	Override R → V table
CONNECTOR_AGENT_RESOURCES	Default resource profile
CONNECTOR_FIRECRACKER_BIN / JAILER_BIN	VMM binaries
CONNECTOR_MICROVM_KERNEL / ROOTFS	Guest assets
CONNECTOR_MICROD_SOCK	microd UDS path
CONNECTOR_PACKET_DNA_REQUIRE	Require DNA on egress
CONNECTOR_WORLD_GRANTS_FAIL_CLOSED	Ungranted deny
CONNECTOR_LLM_DATA_TOKENIZE	Broker tokenize ingress

23. Glossary & module map

Term	Definition
AgentCell	Linux isolation body (ns/cgroup/Landlock/...)
MicroCell	Machine isolation body (Firecracker guest)
CVR	Connector Virtualization Runtime — resolve/bind/probe
microd	Privileged supervisor for VMM ops over UDS
PATE / NF ³	Admit / categorical membrane checks
DIM	Dynamic Intelligence Manifold — cognition, not authority
Knot	Belief field / interference memory plane
BCR	Budget reserve → commit spend protocol
Packet DNA	Outbound effect sealing / lineage
R/A/E	Requested / Applied / Effective posture triad
V0–V4	Isolation profiles from app gates to dedicated MicroCell

Related documents

- `CONNECTOR_CAPABILITY_STANDARD.md` — parent standard
- `CONNECTOR_PRODUCT_PROMISE.md` — promise & anti-claims
- `CONNECTOR_FINAL_OUTCOMES.md` — E/A/S
- `CONNECTOR_AGENT_ISOLATION.md` · `CONNECTOR_ISOLATION_IMPLEMENTATION_PLAN.md`
- `CONNECTOR_OPERATIONAL_SOFTWARE.md` · `CONNECTOR_AUGMENTED_ENV.md`
- `CONNECTOR_REACH_CHECKLIST.md` — verify checklist

End of Architecture Record. This PDF is a durable snapshot of engineered Connector substrate capability for agentic augmented environments. Regenerate from `platform/docs/arch/arch.html` when the stack materially changes.